

De Illis Quae Abstractionem Effugiunt

(On Those Things That Evade Abstraction)

or How Gen Eds Enhanced My Engineering Skills

by: Elliott Bourgeois
Date: 10/7/2026

Courses in engineering tend to reduce problems, solutions, and, by extension, most of the world into quantifiable parameters: voltage, bandwidth, tolerance, efficiency, frequency. While there is an immense usefulness in this reduction (and great simplification of theory based on these assumptions), there is also something important necessarily lost in it. My general education courses have taught me to be suspicious of that quantitative reduction of the world.

German Film Studies (GER 3780) taught me to regard objects and events as products of historical circumstance, cultural memory, and human interpretation. Technological artifacts are much the same, in that they don't emerge from and into some neutral, ahistorical space, but rather an inhabited world, carrying consequences that cannot be apprehended through merely technical analysis. Thinking about engineering in this more holistic way has made me more attentive to the context in which a solution operates, rather than treating the solution itself as the isolated terminus of one singular problem.

Advanced Human Sexuality (HDFS 4760) approached this question from a more intimate direction. In this course, many assumptions we ordinarily make about the human subject were complicated. Individual experience is profoundly mediated by culture, social structures, biology, and personal circumstance. For an engineer, this is an important corrective to numerical/pragmatic abstraction. While it's easy to design for an imaginary, universal user whose needs can be effectively reduced to a fixed set of specifications, actual people are considerably less tractable. Identities, expectations, vulnerabilities, and interpretations differ, and those differences can determine whether a technology is useful, alienating, accessible, or even harmful. Consequently, considerations such as privacy and social consequence are constituent parts of the problem itself, rather than ornamental additions to an engineering solution.

After taking these general education courses, my conception of what constitutes a successful engineering solution has changed drastically. An elegant design may be economically untenable; an efficient system may produce undesirable social consequences; a solution optimized for one population can be inept when transplanted into another cultural or material context. Engineering therefore seems more like an act of negotiation between the idealized abstractions of the engineer and the recalcitrant complexity of the world. My general education courses have given me a greater awareness of and appreciation for that complexity. They have taught me that I should always question what assumptions a product's development and integration entail, whom the product serves, and what consequences accompany its existence.